

■ 3.1.2 Converting Between Types of Numbers

$N[x, n]$	convert x to an approximate real number with at most n digits of precision
$Rationalize[x]$	give a rational number approximation to x
$Rationalize[x, dx]$	give a rational number approximation to within tolerance dx

Functions that convert between different types of numbers.

This gives a 30-digit real number approximation to $3/7$.

```
In[1]:= N[3/7, 30]
Out[1]= 0.428571428571428571428571428571
```

This takes the 30-digit number you have just generated, and reduces it to 20-digit precision.

```
In[2]:= N[%, 20]
Out[2]= 0.42857142857142857143
```

This converts the result back to a rational number.

```
In[3]:= Rationalize[%]
Out[3]=  $\frac{3}{7}$ 
```

The numerical value of π is not “sufficiently close” to a rational number to be converted.

```
In[4]:= Rationalize[ N[Pi] ]
Out[4]= 3.14159
```

If you give a specific tolerance, **Rationalize** will give you a rational number approximation accurate to within that tolerance.

```
In[5]:= Rationalize[ N[Pi], 10^-5 ]
Out[5]=  $\frac{355}{113}$ 
```